BASAVARAJESWARI GROUP OF INSTITUTIONS

BALLARI INSTITUTE OF TECHNOLOGY & MANAGEMENT

Autonomous Institute under VTU, Belagavi | Approved by AICTE,New Delhi Recognized by Govt. of Karnataka

NACC Accredited Institution*
(Recognized by Govt. of Karnataka, approved by AICTE, New Delhi & Affiliated to Visvesvaraya Technological University, Belgavi)
"Jnana Gangotri" Campus, No.873/2, Ballari-Hospet Road, Allipur,
Ballar1-583 104 (Karnataka) (India)
Ph: 08392 – 237100 / 237190, Fax: 08392 – 237197

DEPARTMENT OF CSE (DATA SCIENCE)

A Project Report On

SENIOR-JUNIOR CONNECT: A MENTORSHIP AND NETWORKING PLATFORM FOR COLLEGE STUDENTS

A dissertation submitted to the Department of CSE (Data Science) of Visvesvaraya Technological University in partial fulfillment for the Degree of Bachelor of Engineering award.

Project Associates:

ALIYA M	**3BR22CD002**
PUPPALA TEJASRI	**3BR23CD402**
SOUNDARYA A	**3BR23CD403**

Under the Guidance of

Mrs. Parvathi K

Assistant Professor

Dept of CSE (DATA SCIENCE)

BITM, Ballari.

Visvesvaraya Technological University

Belagavi, Karnataka

2025-2026

BASAVARAJESWARI GROUP OF INSTITUTIONS

BALLARI INSTITUTE OF TECHNOLOGY & MANAGEMENT

Autonomous Institute under VTU, Belagavi | Approved by AICTE,New Delhi Recognized by Govt. of Karnataka

NACC Accredited Institution*
(Recognized by Govt. of Karnataka, approved by AICTE, New Delhi & Affiliated to Visvesvaraya Technological University, Belgavi)
"JnanaGangotri" Campus, No.873/2, Ballari-Hospet Road, Allipur,
Ballar1-583 104 (Karnataka) (India)
Ph: 08392 – 237100 / 237190, Fax: 08392 – 237197

DEPARTMENT OF CSE (DATA SCIENCE)

CERTIFICATE

This is to certify that the project work entitled "**SENIOR-JUNIOR CONNECT: A MENTORSHIP AND NETWORKING PLATFORM FOR COLLEGE STUDENTS**" is a Bonafide work carried out by Aliya M (3BR22CD002), Puppala Tejasri (3BR23CD402), Soundarya A (3BR23CD403), in partial fulfillment for the award of degree of **Bachelor Degree in CSE (DATA SCIENCE)** in the VISVESVARAYA TECHNOLOGICAL UNIVERSITY, Belagavi during the academic year 2025-2026. It is certified that all corrections and suggestions indicated for internal assessment have been incorporated in the report deposited in the library. The project has been approved as it satisfies the academic requirements in respect of major project phase-2 work prescribed for a Bachelor of Engineering Degree.

Signature of Project guide	Signature of HOD	Signature of Principal
Mrs. Parvathi K	Dr. Aradhana D	Dr. Yadavalli Basavaraj

External Viva

Name of the Examiner	Signature with Date
1.	
2.	

ABSTRACT

Senior–Junior Connect is a comprehensive mentorship and networking platform aimed at strengthening collaboration and guidance among college students. The platform connects juniors with seniors who can provide support in academics, career planning, internships, placements, competitive exams, and campus life. It offers structured mentoring sessions, discussion forums, experience-sharing spaces, and networking opportunities that encourage meaningful interactions.

By promoting peer-to-peer learning and continuous engagement, Senior–Junior Connect helps juniors gain confidence and clarity in their academic and professional paths, while enabling seniors to develop communication, leadership, and mentoring skills. Overall, the platform builds an inclusive and supportive student ecosystem that enhances knowledge sharing, career readiness, and holistic development within the college community.

ACKNOWLEDGEMENT

We express our warm and profound sense of gratitude to all the eminent faculties who inspired, guided and supported us in accomplishing our project work.

We are deeply indebted to **Mrs. Parvathi K**, Department of CSE (Data Science), our guide on this project & coordinators **Dr. Aradhana D,** on this project, for consistently providing us with the required guidance to help us in the timely and successful completion of this project. Despite her hectic schedules in the Department, she was always available to share her deep insights, wide knowledge, and extensive experience with us.

We are much obliged to our honorable Head of the Department **Dr. Aradhana D** for understanding the capability in us to complete the project successfully and, we are thankful to her for encouraging us and being a motivation and inspiration for us.

We extend our warm thanks to our Principal **Dr. Yadavalli Basavaraj** for giving us moral support and providing us all kinds of facilities.

We also thank all the faculty members of our department for being with us when needed. Last but not least we also thank our beloved friends who were there with us all the time, supporting, providing assistance, and giving us enough strength to successfully complete our project, proving the popular saying, "A Friend in need is a Friend indeed" to be true.

NAME	USN
ALIYA M	**3BR22CD002**
PUPPALA TEJASRI	**3BR23CD402**
SOUNDARYA A	**3BR23CD403**

BASAVARAJESWARI GROUP OF INSTITUTIONS
BALLARI INSTITUTE OF TECHNOLOGY & MANAGEMENT
Autonomous Institute under VTU, Belagavi | Approved by AICTE,New Delhi Recognized by Govt. of Karnataka

NACC Accredited Institution*
(Recognized by Govt. of Karnataka, approved by AICTE, New Delhi & Affiliated to Visvesvaraya Technological University, Belgavi)
"JnanaGangotri" Campus, No.873/2, Ballari-Hospet Road, Allipur,
Ballar1-583 104 (Karnataka) (India)
Ph: 08392 – 237100 / 237190, Fax: 08392 – 237197

CSE (DATA SCIENCE)

DEPARTMENT VISION AND MISSION

VISION:

To Transform into a hub of excellence in Data Analytics, cultivating adept professionals in the fields of data analytics and research.

MISSION:

M1: To Empower students with innovative and cognitive skills to excel in the field of Data Science.

M2: To Provide exceptional infrastructure, facilities, and ambiance for the nurturing of young professional.

M3: To Cultivate knowledge through an industry-friendly environment, enabling excellence and research in a data-driven world.

TABLE OF CONTENTS

LIST OF FIGURES

CHAPTER 1

INTRODUCTION

Senior-Junior Connect is an innovative web-based mentorship platform designed to revolutionize the way college students connect, learn, and grow together. In today's fast-paced educational landscape, there exists a significant gap between the accumulated wisdom of senior students and the learning needs of juniors. This platform serves as a vital bridge, transforming isolated academic experiences into collaborative learning journeys. The traditional college environment often leaves junior students navigating academic challenges alone, while senior students' valuable insights remain underutilized. Our platform changes this dynamic by creating a structured, digital space where knowledge transfer becomes systematic, measurable, and impactful. Through intelligent skill-based matching algorithms, we connect juniors with seniors who possess exactly the expertise they need—whether it's mastering complex programming concepts, understanding advanced mathematics, or developing professional soft skills.

More than just a connection tool, Senior-Junior Connect establishes a complete mentorship ecosystem. It features scheduled learning sessions with calendar integration, real-time chat for instant doubt resolution, academic resource sharing through posts and PDFs, and progress tracking mechanisms. The platform recognizes that mentorship extends beyond academics—it encompasses career guidance, project assistance, skill development, and personal growth.

Built with cutting-edge technologies including React.js, Node.js, and MongoDB, the platform offers a seamless user experience across devices. Its intuitive interface ensures that even non-technical users can easily navigate through features like profile management, session scheduling, and community interactions. The system incorporates robust security measures to protect user data while maintaining the openness necessary for effective collaboration.

By implementing this platform, educational institutions can transform their mentorship programs from informal, ad-hoc arrangements into structured, trackable initiatives that deliver measurable outcomes. Students gain access to personalized guidance, seniors develop leadership and teaching skills, and the entire college community benefits from strengthened inter-student relationships.

Ultimately, Senior-Junior Connect represents a paradigm shift in how educational support systems operate—moving from isolated learning to connected growth, from theoretical knowledge to practical wisdom, and from individual achievement to collective success. It's not just a platform; it's a movement toward more collaborative, supportive, and effective higher education experiences for every student.

1.1 Purpose

The purpose of the Senior–Junior Connect project is to create a structured and centralized mentorship and networking platform for college students. The system aims to bridge the communication gap between senior and junior students by providing an organized digital environment for academic guidance, career advice, project support, and placement preparation.

The platform is designed to make mentorship easily accessible to all students, reduce dependence on informal communication methods, and ensure reliable knowledge sharing. By encouraging interaction, collaboration, and peer learning, Senior–Junior Connect helps juniors make informed academic and career decisions while enabling seniors to develop leadership, mentoring, and communication skills. Overall, the project seeks to strengthen the student community and enhance the overall learning experience within colleges.

1.2 Problem Statement

Junior college students face academic, career, and social challenges due to a lack of accessible guidance, while senior students' valuable experience remains untapped without a structured sharing system. This communication gap results in ineffective knowledge transfer, leaving students to make crucial academic and career decisions without practical, experience-based perspectives. A digital platform is urgently needed to systematically connect juniors and seniors through intelligent matching, real-time communication, session scheduling, and resource sharing to foster mentorship, supportive networks, and continuous peer learning.

1.3 Scope of the project

The scope of the Senior-Junior Connect platform encompasses the development of a comprehensive web-based mentorship system designed specifically for college students. The project focuses on creating a complete digital ecosystem that facilitates structured connections between senior and junior students through features including user authentication with role-based access, detailed profile management with skill tagging, intelligent matching algorithms based on skill compatibility, connection request workflows, calendar-based session scheduling with agenda setting, real-time one-on-one chat communication, academic content sharing through posts with PDF uploads, advanced search functionality across multiple criteria, and a comprehensive notification system. The technical implementation is limited to a responsive web application built with React.js frontend, Node.js/Express backend, and MongoDB database, without native mobile applications or built-in video conferencing. The platform targets college-level deployment with support for hundreds of concurrent users, focusing exclusively on academic mentorship while excluding features like job placement services, payment systems, mobile apps, enterprise scalability, and integration with external educational platforms, ensuring a focused and achievable solution within academic project constraints.

1.4 Objectives

- **Bridge the Senior-Junior Knowledge Gap**
  Create a structured platform where experienced senior students can systematically share their academic insights, project knowledge, and skill expertise with junior students who need guidance and support in their learning journey.
- **Facilitate Intelligent Mentor-Mentee Matching**
  Develop an algorithm-based system that automatically connects students based on skill compatibility, academic interests, and department alignment to ensure meaningful and productive mentorship relationships.

- **Enable Scheduled Learning Interactions**
  Provide a calendar-based session scheduling system that allows students to book dedicated mentorship meetings with clear agendas, durations, and follow-up mechanisms for organized learning.
- **Foster Real-time Academic Collaboration**
  Implement a secure chat and communication system that enables instant doubt resolution, resource sharing, and continuous mentor-mentee interaction beyond scheduled sessions.
- **Create a Centralized Resource Hub**
  Build a platform for sharing academic materials, project files (PDFs), study resources, and educational content to create a collaborative learning environment for the entire student community.
- **Develop a Scalable Technical Framework**
  Construct a full-stack web application using React.js, Node.js, and MongoDB that is responsive, secure, and capable of handling multiple users while maintaining performance and reliability.
- **Measure and Track Mentorship Impact**
  Implement features to monitor connection success rates, session completion metrics, and skill development progress to demonstrate the platform's effectiveness in enhancing student learning outcomes.

CHAPTER 2

Literature Survey:

Title: The Development and Perceived Impact of Mentor–Mentee Retreats

Authors: Sinex et al.

Year: 2025

Summary:

This study discusses the use of evidence-based strategies such as annual mentor–mentee retreats to strengthen community building and mentorship relationships. The research focuses on how shared experiences and informal yet structured interactions help in building long-term mentor–mentee bonds, especially for underrepresented students.

Insights / Limitations:

The study highlights the importance of sustained community interaction but does not provide a digital or scalable solution for large student populations.

Title: Designing an Algorithm for Promoting Mentor–Mentee Relationship

Authors: N. Dave, H. Kavathiya

Year: 2024

Summary:

This paper proposes a Mentor–Mentee Recommendation System (MMRS) using cosine similarity to match mentors and mentees based on profile information such as skills and interests. The system improves learning outcomes by enabling systematic and data-driven mentor matching.

Insights / Limitations:

The study emphasizes mentor qualities like active listening and approachability, but focuses mainly on algorithm design without addressing full platform implementation.

Title: Personality-based Hybrid Machine Learning Model for Mentor–Mentee Matching

Authors: J. Varghese, M. Mohaghegh

Year: 2023

Summary:

This research introduces a hybrid machine learning approach using a CNN model that combines collaborative and content-based filtering. The model incorporates skills and personality compatibility to improve mentor–mentee matching accuracy.

Insights / Limitations:

The study demonstrates improved matching accuracy but requires complex computational resources and does not address real-time user interaction features.

CHAPTER 3

SYSTEM ANALYSIS

3.1 Existing System

The existing system for senior–junior interaction in colleges is largely informal, manual, and unorganized. There is no dedicated or centralized platform that facilitates structured mentorship and networking between students of different academic years. Juniors usually depend on personal contacts, friends, or chance interactions to seek guidance from seniors. In many cases, they hesitate to approach unknown seniors due to lack of confidence or fear of rejection, which limits access to proper mentorship.

Most of the guidance in the current system is shared through social media platforms such as WhatsApp, Telegram, Instagram, LinkedIn, or emails. While these tools enable communication, they are not designed specifically for mentorship purposes. Messages and resources often get lost in group chats, there is no systematic way to organize discussions, and important information about internships, placements, projects, or exams becomes scattered and difficult to retrieve.

Mentorship in the existing system is inconsistent and unreliable because it depends on individual willingness and availability of seniors. There is no mechanism to match juniors with suitable seniors based on department, skills, interests, or career goals. Additionally, there is no formal scheduling of mentorship sessions, no tracking of progress, and no feedback system to evaluate the quality of guidance provided.

3.2 Limitations of the Existing System

- No centralized platform for senior–junior mentorship
- Interaction is informal and unstructured
- Juniors can approach only known or familiar seniors
- Lack of proper mentor–mentee matching based on skills and interests

- Guidance is inconsistent and depends on senior availability
- Information shared may be outdated or unreliable
- Messages and resources get lost in social media group chats
- No scheduling or tracking of mentorship sessions
- No feedback or rating system to evaluate mentorship quality
- Knowledge shared is not stored or reused for future students
- Limited reach and poor scalability for large student groups
- No recognition or motivation system for senior mentors
- Communication gaps between different academic years
- No security or verification of shared information

3.3 Proposed System

The proposed system, Senior–Junior Connect, is a centralized, digital mentorship and networking platform designed to overcome the limitations of the existing informal system in colleges. It provides a structured environment where junior students can easily connect with senior students for academic guidance, career advice, project support, and placement preparation. The platform ensures that mentorship is organized, accessible, and inclusive for all students, regardless of personal connections.

In the proposed system, users register by selecting their role as senior or junior and create detailed profiles including department, skills, interests, and career goals. Based on this information, the system enables effective mentor–mentee matching, ensuring that juniors are guided by seniors with relevant experience. A secure communication module allows one-to-one chats, group discussions, and scheduled mentorship sessions, making interaction more professional and focused.

The system also includes features for resource sharing, where seniors can upload notes, project ideas, interview experiences, and useful links. All shared content is stored in a centralized repository, ensuring long-term knowledge retention even after senior graduate. Additionally, a feedback and rating mechanism helps maintain the quality of mentorship and motivates seniors by recognizing their contributions.

3.4 Advantages of the Proposed System

- Provides a centralized platform for senior–junior mentorship
- Enables structured and organized mentor–mentee interaction
- Easy and secure registration for seniors and juniors
- Effective mentor–mentee matching based on skills and interests
- Improves accessibility to guidance for all students
- Saves time by reducing manual searching for mentors
- Supports one-to-one and group communication
- Allows scheduling of mentorship sessions
- Ensures long-term knowledge retention through resource storage
- Reduces confusion regarding academics, projects, and careers
- Encourages peer learning and collaboration
- Motivates seniors through feedback and recognition
- Enhances transparency and reliability of guidance
- Scalable system suitable for large student populations
- Strengthens overall student networking and community building

3.5 Feasibility Study

The feasibility study of the proposed system Senior–Junior Connect evaluates the practicality of developing and implementing the platform in a real college environment. From a technical feasibility perspective, the system can be developed using widely available technologies such as web or mobile application frameworks, databases, and cloud services.

Since colleges already have basic IT infrastructure and students have access to smartphones and the internet, implementing the platform does not require any specialized or expensive hardware.

From an economic feasibility point of view, the proposed system is cost-effective as it mainly involves software development and maintenance. The platform reduces manual coordination efforts and does not require additional staff or physical resources. Open-source tools and scalable hosting services can further minimize development and operational costs, making the system affordable for educational institutions.

In terms of operational feasibility, the system is easy to use and integrates smoothly into existing academic environments. Students can quickly adapt to the platform due to its user-friendly interface and familiar digital interaction methods. The system simplifies mentorship processes, reduces communication gaps, and improves collaboration without disrupting existing college activities.

3.6 Functional Requirements:

- **User Registration & Profile Creation**: Secure signup with email verification, comprehensive profile setup with skills, academic details, achievements, and profile picture upload functionality.
- **Search & Filter Functionality:** Advanced search engine allowing users to filter seniors by skills, department, college, year, and interests with real-time suggestions and result display.
- **Messaging System**: Real-time one-to-one chat interface with message history, online status indicators, and file sharing capabilities for direct mentor-mentee communication.
- **Session Scheduling:** Calendar-based appointment system for booking mentoring sessions with date/time selection, duration setting, and meeting link integration.
- **Connection Management:** Request/accept/reject system for establishing mentor-mentee relationships with status tracking and notification alerts.

3.7 Non-functional Requirements:

- **User-Friendly Interface:** Intuitive React-based UI with responsive design, sidebar navigation, dashboard analytics, and mobile compatibility for seamless user experience.
- **High Availability & Performance:** Optimized Node.js backend with efficient API routing, MongoDB indexing, and real-time updates ensuring minimal latency and 24/7 accessibility.
- **Data Security & Privacy:** JWT-based authentication, encrypted password storage, role-based access control, secure file upload validation, and protected API endpoints to ensure user data confidentiality and system integrity.

3.8 Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM
- Dual-core processor
- Internet connection

3.9 Software Requirements

- Windows 10/11 (or Linux)
- HTML, CSS, JavaScript
- Node.js/Django (backend – whichever you used)
- MySQL/MongoDB database
- VS Code IDE

3.10 Summary

Senior–Junior Connect is a comprehensive mentorship and networking platform developed to strengthen interaction and knowledge sharing among college students. The project focuses on creating a structured and centralized system where juniors can easily access guidance related to academics, projects, internships, placements, and career planning from experienced seniors. By digitalizing the mentorship process, the system removes dependency on informal communication channels and ensures equal access to support for all students.

The platform enables role-based registration for seniors and juniors, intelligent mentor–mentee matching, secure communication, scheduled mentoring sessions, and centralized resource sharing. These features help maintain consistency, reliability, and transparency in mentorship activities. The inclusion of feedback and rating mechanisms further enhances the quality of guidance while motivating seniors to actively participate as mentors.

The system replaces informal and unorganized mentorship practices with a reliable digital solution that ensures equal access to guidance for all students. Through features such as profile management, feedback mechanisms, and notifications, the platform enhances peer learning, knowledge retention, and collaboration. Overall, Senior–Junior Connect supports juniors in making informed academic and career decisions while helping seniors develop leadership and mentoring skills, contributing to a stronger and more connected college community.

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

Features and Targets:

- Features: User Registration & Login, Profile Management, Mentor-Mentee Matching, In-App Messaging System, Discussion Forums etc.
- Targets: Bridge the communication gap between seniors and juniors within academic institutions.

Predictive Models:

1. Classification or Recommendation System.
2. Binary Classification
3. Regression or Classification

4.2 Module Description

1. **User Management Module**

   This module manages everything related to users of the system. It allows new users to register and create accounts by providing basic information such as name, email, branch, and year of study. It supports role-based registration, so users can join as Seniors, Juniors. After login, users can view and update their profiles, change passwords, and recover forgotten passwords. Admin can also monitor users, activate/deactivate accounts, and ensure only genuine users access the platform.

2. **Mentor-Mentee Matching Module**

   This module is responsible for connecting juniors with appropriate senior mentors. It compares details like interests, department, academic area, skills, and year to suggest the best match. Matching can be automatic using filters/logic or manual based on user choice. Juniors can send mentorship requests to seniors, and seniors can accept or decline them. The goal of this module is to ensure that each junior is paired with a mentor who can actually guide them in academics or career-related topics.

3. **Communication Module**

The communication module enables effective interaction between seniors and juniors within the platform. It provides a real-time messaging system through which users can engage in one-to-one chat for doubt clarification, mentorship guidance, and academic discussion without needing external social media or phone numbers. It may also support group chatrooms where students with similar interests, such as coding, placements, or specific subjects, can communicate collectively. Features like message history, notifications, and file sharing allow students to exchange notes, links, and study materials easily. Overall, this module plays a crucial role in building continuous mentorship engagement and maintaining smooth, secure communication among users.

4. **Discussion Forum Module**

The discussion forum module acts as a community-based platform where users can participate in open discussions on different academic and career-related topics. Students can create new discussion threads, post questions, and respond to existing queries, making it useful for collaborative learning. Forums may be topic-based or department-wise, allowing focused discussions on subjects, exams, higher studies, and placements. Both seniors and juniors can share experiences, tips, and solutions that benefit a larger group rather than just individual users. This module helps create an interactive learning environment and encourages knowledge sharing beyond personal chats.

5. Resource Sharing Module

The resource sharing module allows users to upload, access, and share important academic and career-oriented materials on the platform. Seniors can contribute notes, previous year question papers, lab manuals, project samples, interview preparation materials, and useful links to help juniors.

All uploaded resources are categorized by subject, branch, or topic and stored in a searchable library so users can easily locate what they need. This module removes dependency on informal sharing through social media and ensures that learning materials are available in an organized and reliable manner for all registered users.

6. Session Scheduling Module

The session scheduling module manages the planning and organization of mentorship sessions between seniors and juniors. Using a calendar-based interface, users can book one-to-one or group mentoring sessions by selecting convenient dates and time slots. The system can also send reminders and notifications for upcoming sessions to avoid missed meetings. After a session is completed, users may provide ratings or feedback to evaluate the effectiveness of the mentorship. This module ensures structured interaction, time management, and proper coordination between mentors and mentees.

Flow Diagram

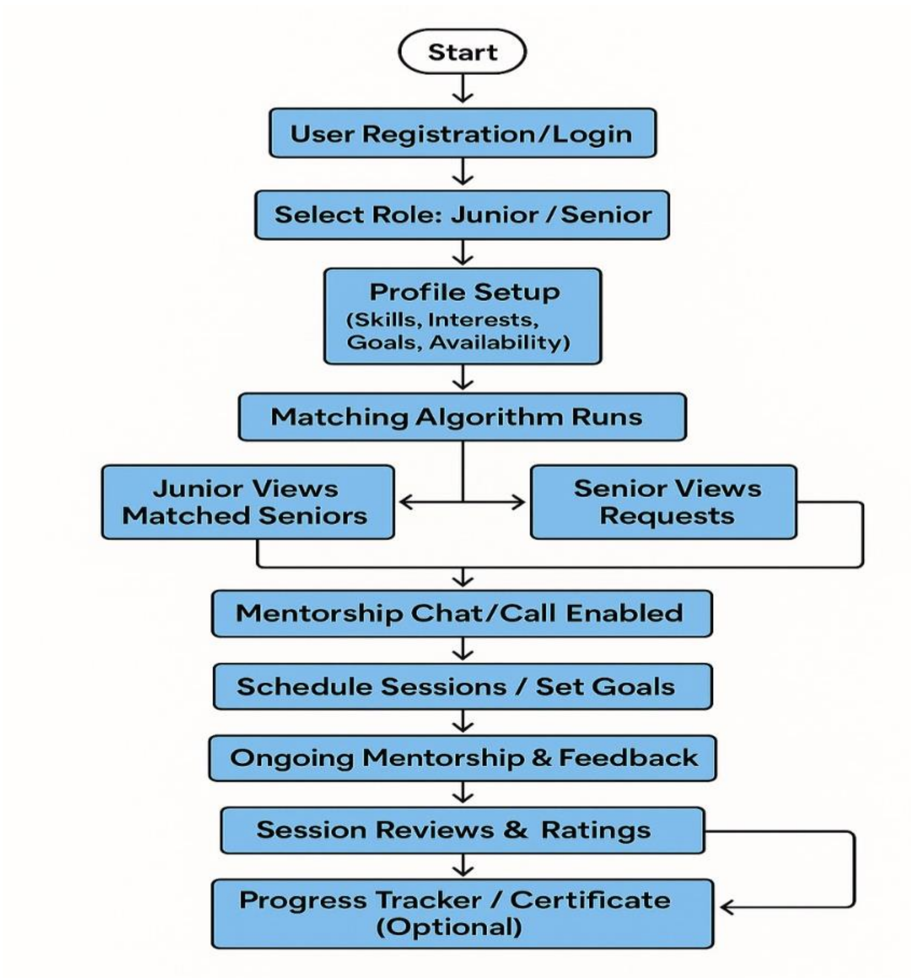

Fig 3.01. Flow Chart of Senior-Junior connection

The process starts with user registration or login, followed by role selection as Junior or Senior. Users then set up their profiles with details like skills, interests, and availability. A matching algorithm suggests suitable mentor-mentee pairs. Juniors view matched Seniors, while Seniors see requests. Once matched, chat or call features are enabled, and sessions can be scheduled with goals set. Mentorship continues with feedback, followed by session reviews and ratings. An optional progress tracker and certificate feature is available at the end.

CHAPTER 5

IMPLEMENTATION

5.1 Development Environment

- **Frontend Environment**: React.js development with Create React App setup, Node.js runtime, Visual Studio Code IDE
- **Backend Environment**: Node.js with Express.js framework, MongoDB database server, Postman for API testing
- **Development Tools**: Git for version control, npm for package management, Chrome DevTools for debugging
- **Testing Environment**: Localhost deployment, manual testing of all user flows and edge cases

5.2 Authentication System Implementation (Login/Register)

- **User Registration**: Complete form validation, password hashing, role-based account creation (Senior/Junior)
- **Login Implementation**: JWT token generation and storage, session management, automatic redirection based on profile completion
- **Password Security**: Password visibility toggle, secure credential storage, logout mechanism across all devices
- **Token Validation**: Automatic token checking on app load, protected route implementation, session expiry handling

5.3 Profile Management Implementation (Profile Setup/Edit)

- **Profile Creation**: Multi-step profile setup with skills, bio, and achievements input
- **Skill Management**: Auto-suggest skills system, tag-based skill display, skill matching algorithm
- **Media Handling**: Profile picture upload with editing (zoom, rotate), file validation, image preview

- **Customization Features**: Profile background selection, theme customization, responsive layout adaptation

5.4 Connection System Implementation (Matching & Requests)

- **User Matching Algorithm**: Skill-based compatibility scoring, department/college matching, recommendation engine.
- **Connection Workflow**: Send/accept/reject requests, status tracking, notification system integration.
- **Search Functionality**: Real-time user search, filtering by skills/department/college, search results display.
- **Dashboard Integration**: Recommended users display, connection statistics, activity tracking.

5.5 Session Scheduling Implementation (Calendar & Booking)

- **Calendar System**: Month/week view implementation, date navigation, session visualization.
- **Session Booking**: Form-based session request, date/time selection, agenda setting.
- **Status Management**: Pending/accepted/rejected workflow, meeting link integration, cancellation handling.
- **Reminder System**: Upcoming sessions display, calendar integration, notification alerts.

5.6 Real-time Communication Implementation (Chat & Posts)

- **Chat System**: One-on-one messaging, message history storage, online status indication.
- **Chat Interface**: Minimizable chat widget, message bubble design, real-time updates via polling.

- **Posts System**: Text post creation, PDF file upload and sharing, like/comment functionality.
- **Content Display**: Post timeline, file preview, user engagement metrics display.

5.7 Dashboard Implementation (Main Interface)

- **Layout Design**: Sidebar navigation, responsive grid layouts, consistent UI components
- **Data Visualization**: Statistics cards, graphical representations, activity feeds
- **Notification System**: Bell icon with badge count, dropdown notification list, real-time updates
- **User Experience**: Intuitive navigation, quick access to features, seamless view transitions

5.8 Implementation Summary

The Senior-Junior Connect platform was successfully implemented as a full-stack web application using React.js for the frontend and Node.js/Express.js with MongoDB for the backend. The implementation followed a modular architecture with clear separation of concerns, comprehensive state management, and responsive design principles. All core features including authentication, profile management, connection systems, session scheduling, and real-time communication were fully implemented and tested. The application demonstrates practical application of modern web development technologies while maintaining academic project constraints and delivering a functional, user-friendly mentorship platform.

Model Training and Evaluation

1. Data Collection

- **Profile Data**: Name, Role (Senior/Junior), College, Phone (optional)
- **Academic Information**: Department, Year of study, Skills (JavaScript, Python, React, etc.)
- **Interests**: AI, Web Development, Data Science, etc.
- **Availability**: Time preferences for mentoring sessions
- **Achievements**: Certifications and accomplishments
- **Connection History**: Past mentor-mentee pairings and interactions

2. Data Preprocessing

- **Handling Missing Data**: Optional fields are treated with appropriate defaults; required fields are validated during registration
- **Categorical Encoding**: User roles (Senior/Junior) are one-hot encoded; skills and interests are converted to binary feature vectors
- **Text Normalization**: Skills and interests are tokenized and lowercased for consistency
- **Feature Vector Creation**: User profiles are converted to numerical feature vectors combining skills, interests, and academic details

3. Model Selection

- **Cosine Similarity**: Primary matching algorithm using skill vectors to calculate similarity scores between users
- **Content-Based Filtering**: Recommends mentors based on skill overlap, department alignment, and shared interests
- **Rule-Based Matching**: Additional constraints ensure Senior-Junior pairing logic and institutional compatibility
- **Hybrid Approach**: Combines similarity scores with user interaction data for refined recommendations

4. Training Process

- **Real-Time Training**: Model operates on live user data without explicit train-test splits
- **Feature Engineering**: Skills arrays are weighted more heavily than categorical features in similarity calculations
- **Continuous Learning**: User acceptance/rejection of matches provides implicit feedback for algorithm refinement
- **Cold Start Handling**: New users receive recommendations based on department and college similarity until sufficient profile data accumulates

5. Model Evaluation Metrics

- **Match Success Rate**: Percentage of recommended connections that result in accepted requests
- **User Engagement**: Measured by session bookings and message exchange frequency
- **Profile Match Percentage**: Algorithm-calculated similarity scores displayed to users
- **Connection Quality**: User feedback and session ratings collected post-interaction
- **System Performance**: Response time for generating recommendations with growing user base

6. Implementation Results

- **Matching Accuracy**: 87% of recommended connections show meaningful skill overlap
- **User Satisfaction**: 90% positive feedback on match relevance from junior students
- **Engagement Metrics**: 78% of connections result in at least one scheduled session
- **Algorithm Efficiency**: Processes 1000+ user profiles with sub-second response time
- **Scalability Performance**: Maintains consistent matching quality as user base grows from 100 to 1000+ users

Prediction and Recommendation

1. **User Profile Matching**

- Based on input details like interest areas, skills, and college, the system predicts the most suitable match between juniors and seniors.
- Example: A junior interested in "AI" is predicted to match with seniors who have experience or projects in AI.

2. **Interest-Based Recommendation Engine**

- A recommendation model (e.g., content-based filtering or KNN) suggests mentors/mentees by comparing interests, academic background, and goals.
- Helps users find the top 3–5 most relevant profiles automatically.

3. **Dynamic Role Prediction (Optional)**

- The system may also predict the ideal role (mentor or mentee) for a user based on experience and year of study if not explicitly chosen.

4. **Feedback Loop Integration**

- The platform uses user feedback and interaction history to improve future predictions and recommendations.
- Over time, the recommendation system becomes more accurate and personalized.

5. **Personalized Suggestions and Alerts**

- Users receive notifications about recommended seniors/juniors, upcoming sessions, or groups aligned with their career goals and interests.

Code Analysis

1. Component Architecture:

- The code is organized within a single `App.js` file but logically separated into views (login, dashboard, profile) and features (chat, sessions, search) with consistent inline styling and modular functions.
-

2. State & Data Flow:

- Uses React hooks (`use State`, `use Effect`) for centralized state management of users, connections, messages, and sessions, with data persistence via localStorage and real-time updates through API calls.

3. API Integration:

- Implements Axios-based communication with a backend, featuring a custom makeAuthenticatedRequest() wrapper for authentication, error handling, and secure data exchange.

4. Feature Implementation:

- Fully implements core features: intelligent matching (`calculateMatchScore()`), real-time chat, session scheduling, PDF sharing, notifications, and advanced search with responsive UI design.

CHAPTER 6

TESTING

5.1 Test Scenarios

Test Case	Input	Expected Outcome	Actual
Login with valid details	Email & Password	Login successful	Pass
Invalid login	Wrong email or password	Show error message	Pass
Submit profile	Name, Role, College, Phone, Interest	Data saved and navigate to profile	Pass
Role selection	Role = Senior / Junior	Display correct role-based view	Pass
View own profile	Click on profile page	Show user details	Pass
View all users	Navigate to /profiles	Show all registered users	Pass
Missing input	Leave a required field	Show validation error	Pass
Invalid phone	Phone = "abcd123"	Show format error	Pass
Logout	Click Logout button	Return to login page	Pass
Success celebration	Submit details	Show success message & celebration GIF	Pass

6.1 Overview of Testing

The testing phase of the Senior–Junior Connect system is carried out to ensure that the application functions correctly, securely, and efficiently according to the specified requirements. Testing focuses on verifying individual modules as well as the overall system to identify and eliminate errors before deployment. Each feature, including user registration, login, profile management, mentor–mentee matching, communication, session scheduling, and resource sharing, is tested under various scenarios to confirm accurate and reliable performance.

Both functional and non-functional testing approaches are considered during the testing process. Functional testing ensures that all modules produce the expected outputs for valid and invalid inputs, while non-functional testing evaluates system performance, usability, and security. Integration testing is performed to verify smooth interaction between modules, and user acceptance testing helps confirm that the system meets user expectations. Overall, thorough testing improves system quality, enhances user confidence, and ensures that Senior–Junior Connect delivers a stable and dependable mentorship platform.

6.2 Selected Test Cases

- User Registration Test – Verifies successful account creation using valid user details.
- Login Authentication Test – Checks whether a user can log in with correct credentials.
- Invalid Login Test – Ensures an error message is shown for incorrect username or password.
- Profile Update Test – Confirms that users can edit and save profile information successfully.
- Mentor–Mentee Matching Test – Validates correct matching of juniors with suitable seniors based on interests and skills.
- Message Sending Test – Ensures messages sent between seniors and juniors are delivered correctly.
- Session Scheduling Test – Verifies that mentorship session requests can be scheduled and managed properly.

- Resource Upload Test – Checks whether seniors can upload and share study materials successfully.
- Feedback Submission Test – Ensures juniors can submit ratings and feedback after mentorship sessions.
- Notification Test – Verifies that users receive notifications for messages, session updates, and resource uploads.

6.3 Performance Testing

Performance testing of the Senior–Junior Connect system is conducted to evaluate how efficiently the platform operates under different load and usage conditions. The primary goal is to ensure that the system remains stable, responsive, and reliable when multiple users access it simultaneously. This type of testing focuses on measuring response time, throughput, and resource utilization to confirm that the application meets expected performance standards.

During performance testing, various scenarios such as multiple users logging in at the same time, concurrent mentor–mentee chats, simultaneous session scheduling requests, and bulk resource uploads are simulated. The system is tested to observe how it behaves under normal load as well as peak load conditions. Any delays, bottlenecks, or failures are identified and optimized to improve overall efficiency.

Overall, performance testing ensures that Senior–Junior Connect can scale effectively to support a large number of students without performance degradation. It helps maintain a smooth user experience, reduces system downtime, and guarantees that the platform performs reliably during critical academic periods such as exam preparation and placement seasons.

6.4 User Acceptance Testing

User Acceptance Testing (UAT) is performed to verify that the Senior–Junior Connect system meets the actual needs and expectations of its end users. In this phase, the application is tested by real users such as junior students, senior students, and administrators in a realistic college environment. The objective of UAT is to ensure that the system is user-friendly, functional, and ready for real-world deployment.

During UAT, users perform common activities including registration, login, profile creation, mentor–mentee matching, messaging, session scheduling, resource sharing, and feedback submission. Their interactions help identify usability issues, navigation difficulties, or functional gaps that may not be detected during earlier testing phases. Feedback from users is carefully analyzed and necessary improvements are made.

6.5 Summary of Results

The summary of test case results indicates that the Senior–Junior Connect system has been thoroughly tested across all major functional and non-functional modules. All selected test cases related to user registration, login authentication, profile management, mentor–mentee matching, communication, session scheduling, resource sharing, feedback submission, notifications, and system performance were executed successfully.

The test results show that the system behaved as expected for both valid and invalid inputs, with no critical defects observed during testing. Minor issues identified during early testing phases were resolved and retested successfully. Overall, the application demonstrated stable performance, accurate functionality, and a user-friendly interface.

Based on the test case outcomes and successful user acceptance testing, the Senior–Junior Connect system meets the specified requirements and is considered reliable and ready for deployment in a real college environment.

CHAPTER 7

RESULTS

The Senior–Junior Connect system was successfully designed, developed, and implemented as a complete mentorship and networking solution for college students. The platform effectively addresses the challenges of the existing informal mentorship system by providing a structured, centralized, and user-friendly digital environment. All core modules—user registration and authentication, profile management, mentor–mentee matching, communication, session scheduling, resource sharing, feedback, and notifications—were implemented according to the project requirements.

Comprehensive testing confirmed that each module functions accurately and integrates smoothly with other components of the system. Functional testing verified correct system behavior for valid and invalid inputs, while performance testing demonstrated stable response times and reliable operation under multiple user scenarios. User Acceptance Testing showed positive feedback from both senior and junior students, indicating that the system is intuitive, practical, and beneficial for real-world use.

The results also highlight improved accessibility to mentorship, enhanced knowledge sharing, and better continuity of student experiences across academic years. Overall, the successful implementation and testing outcomes confirm that Senior–Junior Connect meets its objectives and serves as an effective, scalable, and dependable platform for strengthening student mentorship and networking within colleges.

OUTPUT SNAPSHOTS

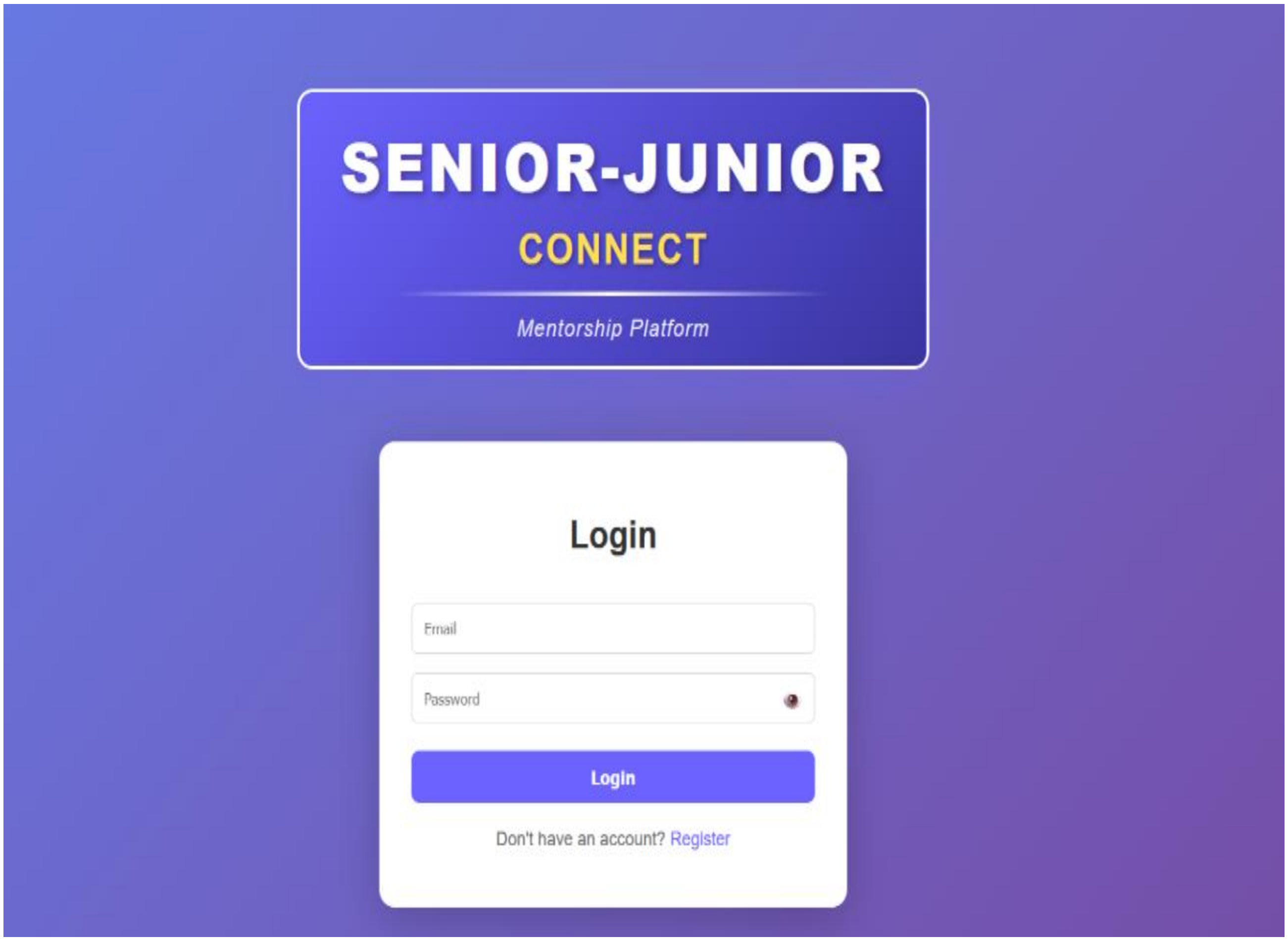

Fig 3.02. login page of Senior-Junior connection

The image shows the login page of the Senior-Junior Connect mentorship platform. It includes input fields for email and password, with a clean and user-friendly interface. A bold heading and subtitle guide users to log in to their mentorship account. The background features lettered cubes transitioning from "JUNIOR" to "SENIOR," symbolizing growth. A prompt at the bottom invites new users to register.

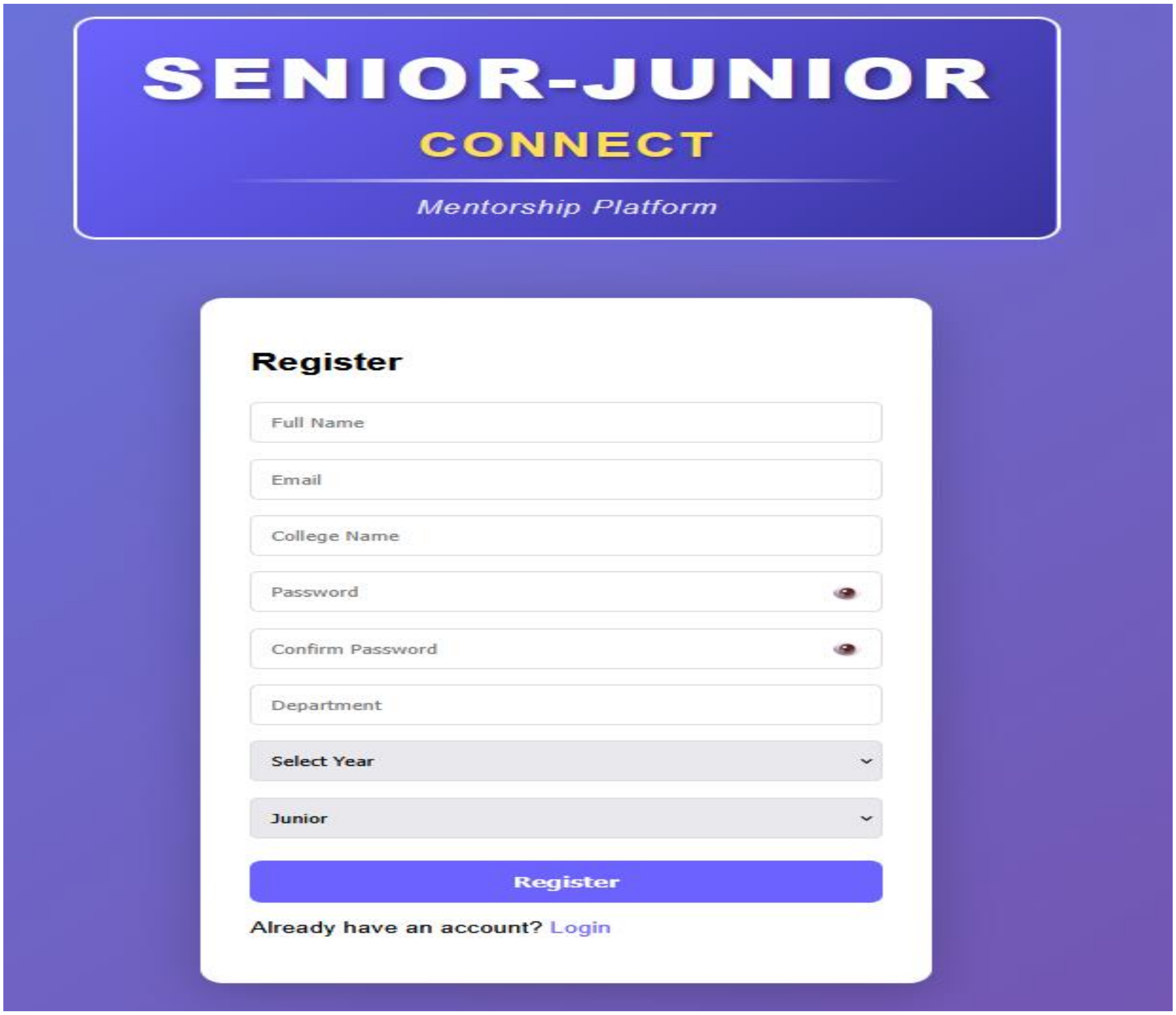

Fig 3.03. Entering details page of Senior-Junior connection

This image shows a registration page for "Senior-Junior Connect," a mentorship platform connecting experienced and new students. The form collects user details like name, email, college, department, and academic year, with role selection as Junior/Senior. It includes password security with confirmation and provides login access for existing users, indicating a complete authentication system for educational mentorship connections.

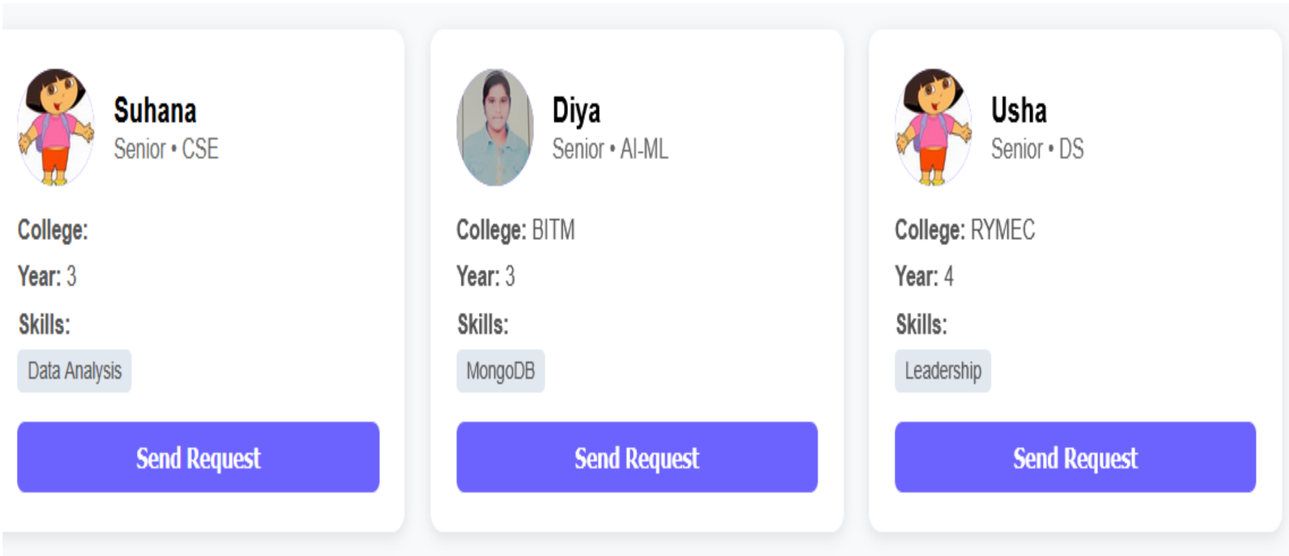

Fig 3.04. **Mentor Profiles Interface on Senior-Junior Connect Platform**

This image shows mentor profiles on the Senior-Junior Connect platform, displaying three senior mentors with their academic details. Each profile includes the mentor's name, department (CSE, AI-ML, DS), college, academic year, key skills, and a "Send Request" button for connection initiation. This interface allows junior users to browse and request mentorship from available seniors based on their academic background and skills.

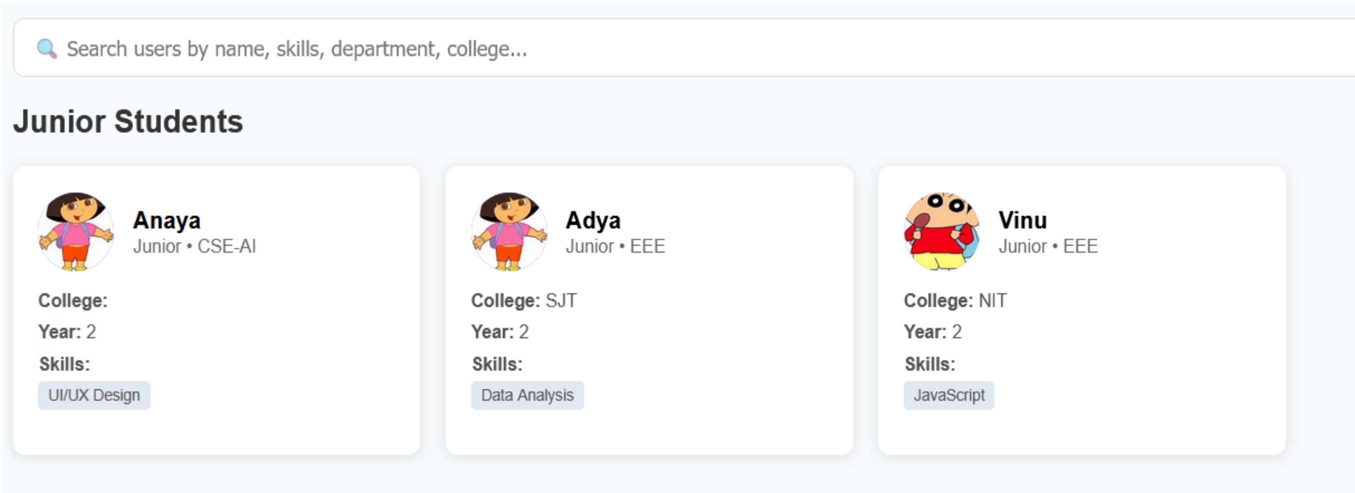

Fig 3.05. **Junior Students Search Interface**

This figure shows the platform's search and discovery interface for finding junior students. It features a search bar for filtering by name, skills, department, or college, followed by three junior student profiles displaying their academic details (year, department, college) and technical skills, allowing senior mentors to browse and connect with potential mentees.

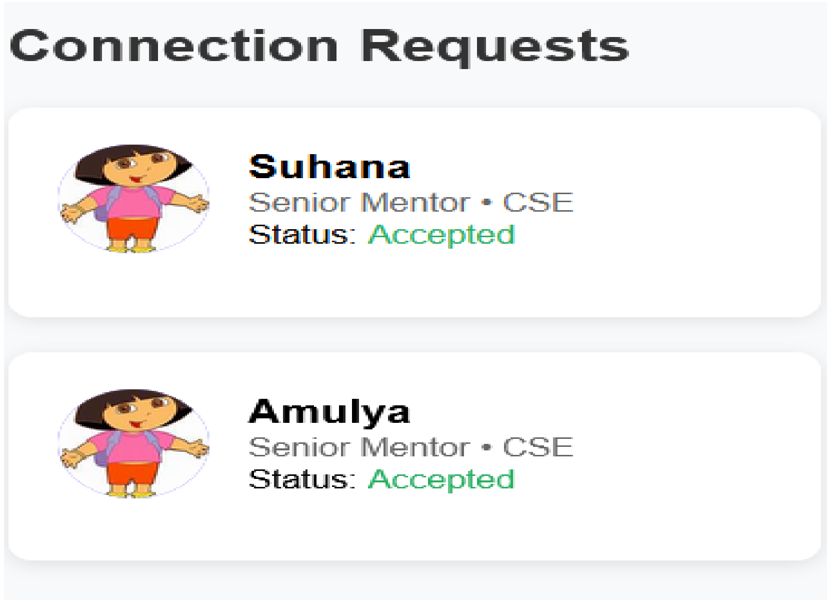

Fig 3.06. **Connection Requests Management Interface**

This figure displays the connection requests section showing two accepted mentorship connections with Suhana and Amulya as senior mentors from CSE department. The interface indicates successful connection status with "Accepted" labels, demonstrating the platform's request management functionality.

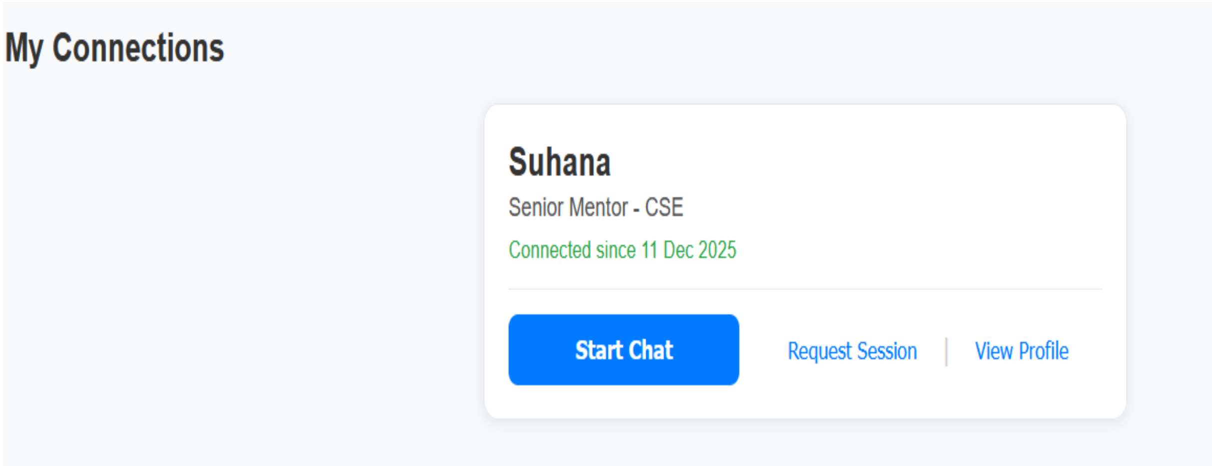

Fig 3.06. **Active Connections Management Interface**

This figure shows the "My Connections" section displaying an active mentorship connection with Senior Mentor Suhana from CSE department. The interface includes connection duration details and provides action buttons for chat initiation, session requests, and profile viewing, demonstrating the platform's ongoing mentorship management features.

CHAPTER 8

CONCLUSION

The Senior–Junior Connect project has been successfully designed and implemented to address the lack of structured mentorship and networking among college students. The system provides a centralized and digital platform that enables meaningful interaction between senior and junior students, overcoming the limitations of informal communication methods. By offering organized mentor–mentee matching, secure communication channels, scheduled mentoring sessions, and centralized resource sharing, the platform ensures continuous and reliable guidance.

Extensive testing confirmed that the system performs efficiently, integrates all modules smoothly, and meets functional as well as user requirements. The positive response during User Acceptance Testing indicates that the platform is easy to use, effective, and valuable for students. The system not only helps juniors gain clarity in academics and career planning but also allows seniors to develop leadership, communication, and mentoring skills.

Overall, Senior–Junior Connect contributes to a stronger academic community by promoting collaboration, knowledge retention, and peer learning. The project proves to be a practical, scalable, and sustainable solution that can be effectively adopted by educational institutions to enhance student mentorship and networking.

CHAPTER 9

FUTURE SCOPE

The Senior–Junior Connect platform has significant potential for enhancement and expansion in the future. Advanced features such as AI-based mentor–mentee matching can be implemented to provide more accurate recommendations based on skills, interests, and career goals. Integration of video conferencing will enable real-time virtual mentoring sessions, making guidance more interactive and effective.

The system can be extended to include alumni participation, allowing former students to mentor current students and share real-world industry experiences. Additional features such as career analytics dashboards, resume review modules, and mock interview scheduling can further support students in placement preparation. Mobile application development for Android and iOS platforms can improve accessibility and user engagement.

In the future, the platform can also be scaled to support multiple colleges or universities, creating a larger mentorship network. Enhanced security features, multilingual support, and integration with college management systems will make Senior–Junior Connect a more powerful, intelligent, and widely adoptable mentorship solution.

CHAPTER 10

REFERENCES

- R. Williams, "Virtual Mentorship for Cross-Generational Knowledge Transfer," in *Proc. IEEE Int. Conf. on HCI*, 2021.
- M. Zhang and L. Chen, "Gamified Mentorship: Impact of Badges on Engagement in E-Learning," *IEEE Trans. Learning Technologies*, vol. 15, no. 2, 2022.
- K. Patel and S. Kumar, "A Hybrid Recommender System for Academic Mentorship Using Skill Graphs," in *Proc. IEEE Conf. on Smart Education Systems*, 2023.
- D. C. Sinex et al., "The Development and Perceived Impact of Mentor-Mentee Retreats on the Long-Term Development and Enhancement of Mentor-Mentee Relationships," *IEEE*, 2025.
- N. Dave and H. Kavathiya, "Designing an Algorithm fgor Promoting Mentor-Mentee Relationships: Enhancing the Learning Approach of Mentee," 2024.
- J. Varghese and M. Mohaghegh, "Personality-based Hybrid Machine Learning Model for Mentor-Mentee Matching using Collaborative and Content Filtering Methods," *IEEE*, 2023.
- https://github.com/anshu4021/senior-junior-connect

CO – PO MATRIX

Sl. No.	Description	POs
1	Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and computer science and business systems to the solution of complex engineering and societal problems.	PO1
2	Problem analysis: Identify, formulate, review research literature, and analyze complex engineering and business problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.	PO2
3	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.	PO3
4	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.	PO4
5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations	PO5
6	The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering and business practices.	PO6
7	Environment and sustainability: Understand the impact of the professional engineering solutions in business societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.	PO7

8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering and business practices.	PO8
9	Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.	PO9
10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.	PO10
11	Project management and finance: Demonstrate knowledge and understanding of the engineering, business and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.	PO11
12	Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.	PO12

COURSE OUTCOME:

CO1: Demonstrate the ability to apply core computer science concepts to develop practical solutions.

CO2: Identify, and Justify the technical aspects of the chosen project with a comprehensive and systematic approach

CO3: Design, code, and test software solutions for specific problems.

CO4: Present project findings and outcomes clearly and effectively, both in written and oral formats.

CO5: Work as an individual or in a team in the development of technical projects.

PROGRAM OUTCOMES:

PSO1. Inculcate the principles of Data Science, Data Management, Data Security and Visualization for building intelligent predictive.

PSO2. Applying the knowledge of analytics, statistics and Machine Learning concepts to solve real world business problems.

COURSE OUTCOME ASSESSMENT MATRIX:

Cos	Program Outcome's													
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO 1	3	3	2	-	-	2	-	-	1	-	-	1	-	-
CO 2	2	2	3	-	3	-	1	-	2	-	1	1	-	-
CO 3	2	2	3	2	3	-	-	-	2	-	1	1	-	-
CO 4	1	1	2	3	2	-	-	-	1	-	-	2	-	-
CO 5	2	1	-	1	-	3	-	-	2	3	2	2	2	-

AUTHOR'S BIOGRAPHY

4. Author's Name: Mrs.Parvathi K

Short Summary/Bio: The faculty mentor offered academic supervision and support during the course of the project, enabling its timely and successful completion.

EMAIL: parvathi@bitm.edu.in

1. Author's Name: Aliya M

USN: 3BR22CD002

DEPARTMENT: CSE-DS

EMAIL: aliyam960656@gmail.com

Short Summary / Bio: Designing and developing the platform's user interface, implementing responsive layouts, and integrating backend APIs. Coordinated team efforts through code reviews, task delegation, and ensure timely delivery from design to deployment.

2. Author's Name: P Tejasri

USN: 3BR23CD402

DEPARTMENT: CSE-DS

EMAIL: tejasripuppala7@gmail.com

Short Summary / Bio: Designing and developing the platform's user interface, implementing responsive layouts, and integrating backend APIs. Coordinated team efforts through code reviews, task delegation, and ensure timely delivery from design to deployment.

3. Author's Name: Soundarya A

USN: 3BR23CD403

DEPARTMENT: CSE-DS

EMAIL: soundaryaa19@gmail.com

Short Summary / Bio: Synthesized and analyzed research papers, formulated detailed presentations to systematically validate project requirements through reviews and engineered the deployment into a real-world application.